

Zone Severity Thresholds and Response Levels

OPERATIONAL — UNRESTRICTED

TEXAS SMART CITY PUBLIC SAFETY PROGRAM
STANDARD OPERATING PROCEDURE

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AUTHORITY: TX-SCPS-POL-001 (AI Surveillance Deployment Policy), TX-SCPS-POL-003 (AI Operations Performance Standards)

Section 1 — PURPOSE

This document defines the per-zone person-count thresholds that determine alert severity levels (SAFE / MEDIUM / HIGH / CRITICAL / EMERGENCY) within the TX-SCPS AI-powered crowd monitoring system. It also specifies the mandatory response actions, Service Level Agreements (SLAs), and responsible parties for each severity level. These thresholds are the authoritative source that justifies why the AI system assigns a given severity rating to an observed crowd density reading.

Section 2 — THRESHOLD DETERMINATION METHODOLOGY

Thresholds are derived from: (a) Physical capacity analysis of each zone (area in square metres / 0.7 m² per person at safe density); (b) Historical incident data and crowd event records; (c) Minimum 6-month review of AI-detected density events against ground-truth officer assessments; (d) Peer consultation with city emergency management and fire marshal occupancy standards.

The AI system produces density_score (0.0–1.0) in addition to person_count. CRITICAL and EMERGENCY alerts require EITHER the person_count threshold OR the density_score threshold to be met, whichever occurs first. This dual-trigger approach prevents large open areas from being under-alerted due to high absolute counts but low spatial density, and prevents small enclosed zones from being under-alerted when absolute counts seem low.

Section 3 — SEVERITY LEVEL DEFINITIONS

SAFE: Normal operations. No crowd management action required.

MEDIUM: Elevated occupancy requiring passive monitoring and logging.

HIGH: Occupancy approaching risk threshold; supervisor notification required.

CRITICAL: Immediate response required; dispatch and EOC notification thresholds apply.

EMERGENCY: Maximum mobilisation; full Zone Response Protocol activated.

Section 4 — ZONE TYPE CLASSIFICATION

All monitored zones are classified into one of five zone types:

- STREET_CORRIDOR: Pavements, pedestrian-only corridors, crossings
- PUBLIC_PLAZA: Open plazas, parks, outdoor gathering spaces
- ENTERTAINMENT: Bars, music venues, high-footfall nightlife streets
- CIVIC_INSTITUTIONAL: Government buildings, convention centres, universities

- TRANSIT_HUB: Bus stops, rail approaches, transit interchange areas
- MARKET_RETAIL: Markets, shopping districts, food vendor areas

Section 5 — PER ZONE TYPE THRESHOLD TABLE

The following table defines severity thresholds by zone type. All person_count values are per camera frame (approximate persons visible in monitored zone field of view). density_score thresholds are secondary triggers as described in Section 2.

STREET_CORRIDOR

Severity	person_count	density_score	Action	Response SLA
SAFE	< 10	< 0.30	No action	—
MEDIUM	10–14	0.30–0.49	Monitor and log	—
HIGH	15–19	0.50–0.69	Notify supervisor	—
CRITICAL	20–29	0.70–0.84	Dispatch + EOC notify	Acknowledge 2 min / Dispatch 5 min / EOC notify 5 min
EMERGENCY	≥ 30	≥ 0.85	Full mobilisation	—

PUBLIC_PLAZA

Severity	person_count	density_score	Action	Response SLA
SAFE	< 20	< 0.25	No action	—
MEDIUM	20–34	0.25–0.44	Monitor and log	—
HIGH	35–49	0.45–0.64	Notify supervisor	—
CRITICAL	50–74	0.65–0.79	Dispatch + EOC notify	Acknowledge 2 min / Dispatch 5 min / EOC notify 8 min
EMERGENCY	≥ 75	≥ 0.80	Full mobilisation	—

ENTERTAINMENT

NOTE: Entertainment zones operate extended CRITICAL sensitivity 22:00–03:00 local time (reduce CRITICAL threshold by 20%).

Severity	person_count	density_score	Action	Response SLA
SAFE	< 15	< 0.30	No action	—
MEDIUM	15–24	0.30–0.49	Monitor and log	—
HIGH	25–39	0.50–0.69	Notify supervisor	—
CRITICAL	40–59	0.70–0.84	Dispatch + EOC notify	Acknowledge 2 min / Dispatch 3 min / EOC notify 5 min
EMERGENCY	≥ 60	≥ 0.85	Full mobilisation	—

CIVIC_INSTITUTIONAL

Severity	person_count	density_score	Action	Response SLA
SAFE	< 30	< 0.20	No action	—
MEDIUM	30–49	0.20–0.39	Monitor and log	—
HIGH	50–74	0.40–0.59	Notify supervisor	—
CRITICAL	75–99	0.60–0.79	Dispatch + EOC notify	Acknowledge 2 min / Dispatch 5 min / EOC notify 10 min
EMERGENCY	≥ 100	≥ 0.80	Full mobilisation	—

TRANSIT_HUB

Severity	person_count	density_score	Action	Response SLA
SAFE	< 25	< 0.30	No action	—
MEDIUM	25–44	0.30–0.49	Monitor and log	—
HIGH	45–64	0.50–0.69	Notify supervisor	—
CRITICAL	65–89	0.70–0.84	Dispatch + EOC notify	Acknowledge 2 min / Dispatch 4 min / EOC notify 6 min
EMERGENCY	≥ 90	≥ 0.85	Full mobilisation	—

MARKET_RETAIL

Severity	person_count	density_score	Action	Response SLA
SAFE	< 20	< 0.25	No action	—
MEDIUM	20–34	0.25–0.44	Monitor and log	—
HIGH	35–54	0.45–0.64	Notify supervisor	—
CRITICAL	55–79	0.65–0.79	Dispatch + EOC notify	Acknowledge 2 min / Dispatch 5 min / EOC notify 8 min
EMERGENCY	≥ 80	≥ 0.80	Full mobilisation	—

Section 6 — PER-LOCATION THRESHOLD OVERRIDES

The following specific camera locations carry threshold overrides based on field analysis. The override value applies in place of the zone-type default for the CRITICAL trigger.

The current system-wide threshold_critical value of 15 persons (set in config/streams_metadata.yaml) is the operational CRITICAL trigger for AI alert generation. This conservative value is intentional for initial deployment and will be calibrated upward as ground-truth data accumulates.

Location-specific notes:

- Texas State Capitol Grounds (stream_id: 3, zone: CIVIC_INSTITUTIONAL): CRITICAL threshold: 15 persons (security-elevated; override applies regardless of zone type). EMERGENCY threshold: 25 persons. Special condition: Any CRITICAL alert during legislative session hours (08:00–18:00 Mon–Fri) requires immediate Capitol Police notification in addition to standard EOC notification.
- 6th Street Entertainment District (stream_id: 4, zone: ENTERTAINMENT): CRITICAL threshold: 15 persons per camera zone (each camera covers a sub-block segment). EMERGENCY threshold: 25

persons. Special condition: Weekend (Fri–Sat) 21:00–02:00 threshold automatically reduced to 12 for heightened sensitivity.

- Austin Convention Center North/South (stream_ids: 6, 7, zone: CIVIC_INSTITUTIONAL): CRITICAL threshold: 15 persons per camera zone. Event mode: When a convention event is registered in the permit system, thresholds increase to 40 CRITICAL / 60 EMERGENCY (event-mode thresholds require OC supervisor activation).
- Dealey Plaza Main (stream_id: 21, zone: PUBLIC_PLAZA): CRITICAL threshold: 15 persons. Historical sensitivity: This location has historical significance; any CRITICAL alert is automatically escalated to MEDIUM-priority EOC notification regardless of duration.
- Congress Avenue Main Plaza (stream_id: 1, zone: PUBLIC_PLAZA): CRITICAL threshold: 15 persons. Event mode: During permitted events, apply event-specific plan on file with zone commander.
- Klyde Warren Park East/West (stream_ids: 22, 23, zone: PUBLIC_PLAZA): CRITICAL threshold: 15 persons per camera zone. Combined zone rule: If BOTH stream 22 and stream 23 are simultaneously CRITICAL, trigger immediate EOC notification regardless of duration (multi-zone rule).
- Whole Foods Market Area (stream_id: 12, zone: MARKET_RETAIL): CRITICAL threshold: 15 persons. Note: Congestion is frequently caused by delivery vehicle access during 06:00–10:00; OC may use judgment to delay dispatch during this window if context is confirmed safe.
- UT Tower Mall / Guadalupe & 24th St (stream_ids: 16, 17, zone: CIVIC_INSTITUTIONAL): CRITICAL threshold: 15 persons. Academic calendar dependency: During semester start/end weeks and graduation events, thresholds may be elevated by OC supervisor with zone commander approval.

Section 7 — RESPONSE ACTIONS BY SEVERITY LEVEL

SAFE: Automated: Dashboard displays green status indicator. Human action: None required. Normal patrol schedule applies.

MEDIUM: Automated: Dashboard displays yellow status indicator. Logged to TimescaleDB. Human action: OC monitoring officer notes reading. No active dispatch. If MEDIUM persists > 30 minutes, log to Incident Log and flag for trend review.

HIGH: Automated: Dashboard displays orange status indicator. Alert visible in Alerts Panel. Human action: OC supervisor notified immediately. Nearest available field unit advised to increase proximity patrol. OC records HIGH event start time in Incident Log. If HIGH persists > 15 minutes without improvement, treat as CRITICAL.

CRITICAL: Automated: Dashboard generates CRITICAL alert. WebSocket push to /ws/alerts. Alert available at GET /api/v1/alerts. Human action: Execute TX-SCPS-SOP-001 Crowd Density Alert Response in full. Mandatory actions: OC supervisor acknowledges alert within 2 minutes. Zone Response Protocol activated. Minimum 2 officers dispatched within 5 minutes. EOC notification if CRITICAL persists > 5 minutes.

EMERGENCY: Automated: Dashboard generates EMERGENCY alert. All monitoring officers receive notification. Human action: Full Zone Response Protocol activation immediately. EOC notification within 2 minutes (mandatory, no duration condition). Sector commander and Watch Commander briefed immediately. All available units in adjacent zones placed on standby. Consider activating mutual aid if person_count continues to rise. Media PIO notified to prepare statement.

Section 8 — THRESHOLD REVIEW AND CALIBRATION PROCESS

8.1 Thresholds are reviewed semi-annually (April and October) or after any EMERGENCY incident.

8.2 Calibration inputs:

- Ground-truth density assessments from field officers filed within 48 hours of CRITICAL events
- AI false-positive / false-negative rate from the monthly accuracy audit (TX-SCPS-POL-003)
- Zone physical capacity data updates from city planning department

8.3 Threshold change requests must be submitted by the zone commander to the Smart City Operations Division.

8.4 Any decrease in a CRITICAL threshold (making it more sensitive) requires OC Director approval.

8.5 Any increase in a CRITICAL threshold (making it less sensitive) requires both OC Director and Privacy Officer approval per TX-SCPS-POL-004 (to ensure civil liberties safeguards remain appropriate).

8.6 All approved threshold changes must be applied to config/streams_metadata.yaml and the AI inference server configuration simultaneously to maintain consistency.

Section 9 — DUAL-TRIGGER RESOLUTION

If person_count and density_score indicate different severity levels:

- Use the HIGHER of the two severity levels as the operative level.
- Document the discrepancy in the Incident Log.
- Report to AI Operations as a potential calibration signal (density model may be detecting crowd packing that the object count does not fully capture, e.g. overlapping persons, partial occlusion).

Section 10 — DOCUMENT CONTROL

Owner: Smart City Operations Division

Approver: Chief of Public Safety Operations

Next Review: 2026-10-01

Related Documents:

- TX-SCPS-POL-001 (AI Surveillance Deployment Policy)
- TX-SCPS-POL-003 (AI Operations Performance Standards)
- TX-SCPS-SOP-001 (Crowd Density Alert Response)
- TX-SCPS-SOP-003 (AI Pipeline Operational Runbook)

END OF DOCUMENT — TX-SCPS-SOP-002 v1.0